ACoP 2026 · From Models to Medicine · Hands-on Exercise 1

Dummy ADC CUI for a first RP2D look

Three endpoints. Seven discrete doses. Compute one CUI by hand before you touch a slider. Local app — not the morning Lilly Shiny.

Walk this deck while running shiny::runApp("scripts/03_app.R")

What this exercise is — and is not

- A first pass through the CUI workflow on a dummy ADC.
- An oncology RP2D-style question: which of 7 doses maximizes utility.
- Multiplicative CUI (the original dummy ADC product).
- Easy on purpose: 3 attributes, discrete doses, one hand calculation.

We will not

- Run rxode2 popPK or brms live.
- Use patient-level trial data.
- Treat a dummy peak as a labeled dose.
- Need five endometriosis endpoints yet — that is Exercise 2.

LEARNING OBJECTIVES

By the end of this block you can

- Name the three attributes and invert the two safety probabilities.
- Normalize a raw weight vector so the $\hat{w}$ sum to 1.
- Compute both the product CUI and the additive U by hand at 5.4 mg/kg.
- Show that changing weights (efficacy vs safety) moves the dummy peak off the label-like dose.
- Say which form you used — product (this exercise) or sum (Exercise 2).

STEP 1

The clinical question

For a dummy folate-receptor ADC, which Q3W mg/kg dose maximizes clinical utility — and does that answer change when you put more weight on response versus on neutropenia and dose reduction?

- Seven discrete doses: 0.8, 1.6, 3.2, 5.4, 6.4, 7.4, 8 mg/kg.
- 5.4 mg/kg is the label-like stand-in (approved MIRV is about 6 mg/kg IBW Q3W).
- This is a population Recommended Phase 2 Dose RP2D-style question.
  Exercise 2 will make it a patient-dose question.
- Same math both times: declare weights, invert safety, score every dose, report argmax.

WHY THIS CASE FIRST

Same CUI workflow — Exercise 2 will change the setting

Axis	Exercise 1 — dummy ADC	Exercise 2 — dummy GnRH antagonist
Decision	Here: population RP2D among 7 doses	Ex 2: which approved dose for this patient
Benefit	ORR (one number)	DYSM, NMPP, DYSP
Risk	Neutropenia, dose reduction	Hot flush, BMD loss
Weights	Expert sliders (you pick them)	DCE patient preferences
CUI math	Product Π $EP^{\hat{w}}$ (default)	Sum Σ $\hat{w} \cdot EP$ (default)
Doses	Discrete mg/kg ladder	10 mg TDD grid, two label doses

STEP 2

Three attributes, one desirability scale

Benefit — already higher is better

- ORR — objective / overall response probability
- Green in the app, solid line.

Risk — invert before scoring

- NoNEU = 1 − P(grade 3+ neutropenia)
- NoDR = 1 − P(dose reduction)
- Orange dashed / purple dotted in the app.

Write this once: every EP that enters the CUI is already on 0–1 and points the same way. If you put raw P(neutropenia) into a positive-weight product, you reward toxicity and the “best” dose runs to 8 mg/kg for the wrong reason.

STEP 3

Dummy rates on the seven-dose ladder

Dose	ORR	G3+ NEU	Dose red.	NoNEU	NoDR
0.8	20%	4%	6%	0.96	0.94
1.6	32%	8%	10%	0.92	0.90
3.2	42%	14%	14%	0.86	0.86
5.4 label-like	**50%**	**25%**	**20%**	**0.75**	**0.80**
6.4	56%	38%	30%	0.62	0.70
7.4	60%	50%	40%	0.50	0.60
8.0	62%	60%	48%	0.40	0.52

ORR keeps rising; neutropenia rises faster after 5.4. That is why efficacy weights walk up the ladder and safety weights walk down.

STEP 4 — WRITE THIS ON THE BOARD

Multiplicative CUI (this exercise)

Original dummy ADC app · geometric index

$$CUI(d) = \prod EP_i(d)^{\hat{w}_i} \qquad \text{where} \quad \hat{w}_i = w_i \,/\, \Sigma_j w_j$$

- w_i are raw importance weights. They do not need to sum to 1 until you normalize.
- EP_ORR = p_ORR. EP_NoNEU = 1 − p_NEU. EP_NoDR = 1 − p_DR.
- If any EP is 0, the product is 0 — a hard veto.
- The dose you report is argmax among the 7 rows, then say whether it is below / at / above 5.4.

STEP 4B

Two CUI forms — say which one you are using

Multiplicative · Exercise 1 default

$$CUI(d) = \prod EP_i(d)^{\hat{w}_i}$$

- $EP_i = 0 \Rightarrow CUI = 0$.
- This is the dummy ADC formula.
- A veto only if a high-weight EP is near zero.

Additive · Exercise 2 default

$$U(d) = \sum \hat{w}_i \cdot EP_i(d)$$

- A weighted average. No single veto.
- Winzenborg 2021 Eq. 1.
- Task D asks you to switch forms on the same weights.

TASK 0 · 8 MINUTES · PENCILS

Compute CUI by hand at 5.4 mg/kg

Given p_ORR = 0.50 p_NEU = 0.25 p_DR = 0.20
Raw w ORR = 2 NoNEU = 1 NoDR = 1

- Step 1 Invert safety. EP_ORR = ____ EP_NoNEU = ____ EP_NoDR = ____
- Step 2 Normalize. ŵ_ORR = ____ ŵ_NoNEU = ____ ŵ_NoDR = ____
- Step 3 Additive. U = ŵ·EP + ŵ·EP + ŵ·EP = _____
- Step 4 Multiplicative. CUI = EP^ŵ × EP^ŵ × EP^ŵ = _____
- Work the arithmetic. The next slide is the key — do not skip ahead.

Same three numbers, both CUI forms

Steps 1–2 shared

EP_ORR = 0.50
EP_NoNEU = 1 − 0.25 = 0.75
EP_NoDR = 1 − 0.20 = 0.80
sum w = 2 + 1 + 1 = 4
ŵ = 2/4, 1/4, 1/4
= 0.50, 0.25, 0.25

Step 3 Additive U = Σ ŵ EP

U = 0.50×0.50 + 0.25×0.75 + 0.25×0.80
= 0.250 + 0.188 + 0.200
= 0.637

Step 4 Multiplicative Π EP^ŵ

0.50^0.50 = 0.707
0.75^0.25 = 0.931
0.80^0.25 = 0.946
CUI = 0.707 × 0.931 × 0.946 = 0.622

LAPTOP

How to run the walkthrough

- Working directory: Hands_On1_20thSept2026/Materials
- Once: source("scripts/01_build_endpoint_grid.R")
- In the room: source("scripts/02_cui_walkthrough.R") or shiny::runApp("scripts/03_app.R")
- Prefer the app. Presets are Task A / B / C. Form starts on Multiplicative.
- Packages: dplyr, tidyr, ggplot2, shiny, bslib, plotly. No brms, no rxode2.
- If R fails, you already finished the important part (Task 0).

TASK A · 5 MINUTES

Balanced weights — original ADC app

w = { ORR = 2, NoNEU = 1, NoDR = 1 } form = multiplicative

- This is ŵ = 0.50 / 0.25 / 0.25 — the weights you just used by hand.
- In the app: choose “Task A — balanced”. Leave the form on Multiplicative.
- Fill in: peak dose = _____ mg/kg at / below / above 5.4 ?
- CUI at 3.2 = _____ CUI at 5.4 = _____ CUI at 7.4 = _____
- Predict: should the dummy like the label-like dose when response is only twice each safety?

TASK B · 5 MINUTES

Efficacy-priority weights

w = { ORR = 8, NoNEU = 1, NoDR = 1 } form = multiplicative

- Same dummy E–R. Only the ORR weight changed.
- Fill in: peak dose = _____ mg/kg did the argmax move? yes / no
- You are paying more for every extra response point, so the ladder can climb.
- Say out loud: the data did not change; the decision-maker did.

TASK C · 5 MINUTES

Safety-priority weights

w = { ORR = 2, NoNEU = 4, NoDR = 4 } form = multiplicative

- Now neutropenia and dose reduction dominate.
- Fill in: peak dose = _____ mg/kg below / at / above 5.4 ?
- Look at 7.4 on the rate table: ORR 60% but G3+ NEU 50%. A safety-first CUI should refuse that.
- Exercise 2 will do the same trick with dyspareunia and bone.

TASK D · 5 MINUTES

Keep Task A weights — switch the form

- Task A weights, Multiplicative then Additive. Fill in both peaks: _____ / _____
- A product is a veto only when an attribute is BOTH low AND heavily weighted.
- Here no EP is near zero, so the two forms stay close — but they do not pick the same row.
- Do not say “multiplicative always picks a lower dose.” Exercise 2 Task D will prove that.
- Write the form in the methods sentence of any CUI you ever ship.

DEBRIEF

The sentence you should be able to say

With balanced weights the dummy ADC CUI peaks at the label-like 5.4 mg/kg; efficacy-priority walks up to 7.4; safety-priority walks down to 3.2. Write the weights and the form (product vs sum) before you report a peak.

- You can now compute a CUI without the app. That was the point of Task 0.
- A dummy peak is a teaching signal, not an RP2D.
- Next: Exercise 2 — same workflow, chronic disease, additive default, patient-level weights.

DO NOT PROJECT UNTIL THE ROOM IS DONE

Answers · Task 0, A, B

Task 0 · 5.4 mg/kg by hand

- Invert 0.50 / 0.75 / 0.80. $\hat{w}$ = 0.50 / 0.25 / 0.25.
- Additive U = 0.637. Multiplicative CUI = 0.622.

Task A · balanced product

- Peak 5.4 mg/kg, CUI 0.622 (matches Task 0). At the label-like dose. CUI 3.2 = 0.601; CUI 7.4 = 0.573.

Task B · efficacy product

- Peak 7.4 mg/kg, CUI 0.589. Argmax moved? YES — above label-like.

DO NOT PROJECT UNTIL THE ROOM IS DONE

Answers · Task C, D

Task C · safety product

- Peak 3.2 mg/kg, CUI 0.745. Below label-like.
- Same dummy E–R as Task B. Opposite direction.

Task D · form switch on Task A weights

- Multiplicative 5.4 mg/kg (0.622). Additive 3.2 mg/kg (0.640).
- Additive at 5.4 is 0.637 — almost tied. The product moves right, to the label-like dose.
- Hand-off: Exercise 2 keeps this contrast and adds five attributes plus DCE weights.

ACoP 2026 · From Models to Medicine · Hands-on Exercise 2

Elagolix-like CUI for endometriosis

Reconstructed from publicly available Phase 3 rates and DCE weights.
Horizon fixed at 6 months. Local app — no external hosting.

Walk this deck while running shiny::runApp("scripts/03_app.R")

What this exercise is — and is not

- A second pass through the same CUI workflow you used on the dummy ADC.
- A chronic, non-oncology decision: which marketed-like dose for which patient.
- Additive multi-attribute utility (Winzenborg 2021 Eq. 1).
- Public rates only. The app runs entirely on your laptop.

We will not

- Fit popPK or Markov E–R live.
- Use patient-level trial data.
- Treat a dummy peak as a prescription.
- Change the 6-month horizon.

LEARNING OBJECTIVES

By the end of 45 minutes you can

- Name the five attributes and say why safety is inverted before it enters the CUI.
- Write the additive CUI equation and contrast it with the Exercise 1 product.
- Compute U(d) on the dummy grid for a declared weight vector.
- Show that changing one weight (dyspareunia) can flip 150 mg QD vs 200 mg BID.
- Explain why a DCE-average peak near 230 mg is a hypothesis, not a label.

STEP 1

The clinical question

For a dummy oral GnRH antagonist in endometriosis-associated pain, which total daily dose maximizes clinical utility — and does that answer change when the patient's preferences change?

- Approved-like regimens on the x-axis: 150 mg once daily (TDD 150) and 200 mg twice daily (TDD 400).
- Duration is fixed at the Phase 3 primary window (about 6 months).
- This is a personalized-dose question, not an MTD / RP2D question.
- Taylor et al. already noted the two doses might be tailored to balance pain vs hypoestrogenic effects.

WHY A SECOND CASE

What changes when you leave oncology

Axis	Exercise 1 — dummy ADC	Exercise 2 — dummy GnRH antagonist
Decision	ADC Exercise 1: population RP2D	Here: which approved dose for this patient
Benefit	ORR (binary)	DYSM, NMPP, DYSP responder rates
Risk	Neutropenia, dose reduction	Hot flush, BMD loss
Weights	Expert sliders	DCE patient preferences (Poulos 2019)
CUI math	Product Π EP^w	Sum Σ $\hat{w} \cdot EP$ (default)
Toxicity tolerance	Higher	Much lower; chronic use

STEP 2

Five attributes, one desirability scale

Benefit — already “higher is better”

- DYSM — dysmenorrhea clinical responder
- NMPP — nonmenstrual pelvic pain responder
- DYSP — dyspareunia responder (secondary; 150 mg was not significant)

Risk — invert before scoring

- NoHF = 1 − P(moderate / severe hot flush)
- BMDok = 1 − P(BMD loss > 1%)
- The 1% BMD cutoff is the Winzenborg surrogate for later fracture concern.

Write this once: every EP_i(d) that enters the CUI is already on 0–1 and points in the same direction. If you forget to invert hot flush or BMD, the “best” dose will run to the toxic tail.

STEP 3

Published Phase 3 anchors we calibrated to

Endpoint	Placebo	150 mg QD	200 mg BID	Source
DYSM responder	20–23%	43–46%	72–76%	Taylor NEJM 2017, mo 3
NMPP responder	37%	50%	55–58%	Taylor NEJM 2017, mo 3
DYSP responder	~31%	~39% NS	~54% sig.	Secondary; 150 mg NS
Hot flush	7%	24%	42%	EM-I, 6 months
Lumbar BMD >5% loss	1.1–1.8%	2.3–3.8%	16–21%	Taylor, month 6

The dummy grid smooths these three anchors with Emax curves on TDD 0–400 mg. CUI uses a 1% BMD-loss surrogate (Winzenborg), which is more sensitive than the published >5% rates.

STEP 4 — WRITE THIS ON THE BOARD

Additive CUI (this exercise)

Winzenborg, Soliman, Shebley · CPT:PSP 2021 · Equation 1

$$U(d) = \sum \hat{w}_i \cdot EP_i(d) \qquad \text{where} \quad \hat{w}_i = w_i \,/\, \sum_j w_j$$

- w_i are raw importance weights (DCE, clinician, or slider). They do not need to sum to 1 until you normalize.
- $EP_i(d)$ is the dummy mean desirability at total daily dose d, already inverted for safety.
- U(d) is a weighted average, so it stays inside [0, 1] if every EP does.
- The dose you report is argmax_d U(d), then the nearest approved-like regimen.

STEP 4B

Invert safety before the sum

Do this

- EP_DYSM(d) = P(responder)
- EP_NMPP(d) = P(responder)
- EP_DYSP(d) = P(responder)
- EP_NoHF(d) = 1 − P(hot flush)
- EP_BMDok(d) = 1 − P(BMD loss > 1%)

Do not do this

- Putting raw P(hot flush) into the sum with a positive weight rewards toxicity.
- Mixing 0–1 probabilities with % BMD change on a different scale.
- Forgetting to say the horizon (we use ~6 months).

STEP 4C

Two CUI forms — say which one you are using

Additive · Exercise 2 default

$$U(d) = \Sigma \ \hat{w}_i \cdot EP_i(d)$$

- One weak attribute cannot veto the dose.
- Matches Poland / Ouellet / Winzenborg.
- Easier to explain as a weighted average.

Multiplicative · Exercise 1 ADC app

$$CUI(d) = \Pi \ EP_i(d)^{\hat{w}_i}$$

- $EP_i = 0 \Rightarrow CUI = 0$ (a hard veto).
- This is the dummy ADC formula.
- Task D asks you to switch forms on the same weights.

DCE average patient (Poulos 2019)

Attribute	Reported weight	In this dummy
Avoid moderate / severe hot flush	3.58 – 3.66	We use 3.62
Improve DYSP	1.70	
Improve NMPP	1.49	
Improve DYSM	1.48	
Avoid later bone-fracture risk	0.49	Mapped to BMDok

Patients put the most weight on avoiding hot flushes, then on the three pain domains, and relatively little on later fracture risk. That is why a DCE-average CUI does not automatically pick the highest efficacy dose.

LAPTOP

How to run the walkthrough

- Working directory: Hands_On2_20thSept2026/Materials
- Once: source("scripts/01_build_endpoint_grid.R") # writes data/endpoint_grid.csv
- In the room: source("scripts/02_cui_walkthrough.R") # prints Tasks A–D and writes figures/
- Optional: shiny::runApp("scripts/03_app.R") # same math, sliders, no internet
- Packages: dplyr, tidyr, ggplot2, shiny, bslib
- If R fails, stay on this deck. Answers are on the last three slides — do not jump there yet.

TASK A · 5 MINUTES

Trial-like weights — DYSP off

w = { DYSM = 1, NMPP = 1, DYSP = 0, NoHF = 1, BMDok = 1 }

- This is the Phase 3 primary-endpoint view: dyspareunia is ignored.
- In the app: choose “Task A — trial-like”. Leave the form on Additive.
- Fill in: peak TDD = _____ mg nearer approved regimen = _____
- CUI at 150 mg = _____ CUI at 400 mg = _____
- Predict before you look: should 200 mg BID win if DYSP is worth nothing?

TASK B · 5 MINUTES

Dyspareunia-priority weights

w = { DYSM = 1, NMPP = 1, DYSP = 2, NoHF = 1, BMDok = 1 }

- Same dummy E–R. Only one weight changed.
- Fill in: peak TDD = _____ mg nearer approved regimen = _____
- Did the argmax move from Task A? yes / no
- This is the Winzenborg Figure 2b lesson: 150 mg QD was not significant for DYSP; 200 mg BID was.
- Say out loud: the data did not change; the decision-maker did.

TASK C · 6 MINUTES

DCE average patient

w = { DYSM = 1.48, NMPP = 1.49, DYSP = 1.70, NoHF = 3.62, BMDok = 0.49 }

- These are published preference weights, not your opinion.
- Fill in: peak TDD = _____ mg (paper reported about 230 mg)
- Is that an approved dose? yes / no
- If you must pick a label dose, which approved-like regimen is closer? _____
- CUI can nominate a hypothetical dose. The label still has only 150 QD and 200 BID.

TASK D · 6 MINUTES

Keep the weights — switch the CUI form

- DCE average, Additive then Multiplicative. Fill in both peaks: _____ / _____
- Now choose Bone-sparing and toggle the form again. Fill in: _____ / _____
- A product is a veto only when an attribute is BOTH low AND heavily weighted.
- DCE barely weights BMD, so the product does not automatically move left.
- Bone-sparing does weight BMD. That is when the two forms can disagree.
- Exercise 1 used the product. Say which form you used in the methods sentence.

IF YOU HAVE 5 EXTRA MINUTES

Two virtual patients (Winzenborg Figure 4)

Pain-dominant

- High weights on DYSM, NMPP, DYSP.
- Low weight on BMD.
- Predict: 200 mg BID.
- Fill in peak TDD = _____

Bone-sparing

- High weight on BMDok.
- Modest weights on pain.
- Predict: 150 mg QD, and a flatter CUI.
- A plateau is a result. Do not force a single number.

DEBRIEF

The sentence you should be able to say

Under trial-like weights the dummy CUI peaks at 150 mg QD; if dyspareunia is twice as important the peak moves to 200 mg BID; DCE-average preferences peak at a non-approved in-between dose. Weights are the model.

- Do not ship a CUI without the weight source, the form (sum vs product), and the horizon.
- A CUI peak that is not an approved dose is a development signal, not a prescription.
- Next: your own indication — write the attributes and a first weight vector before you fit E–R.

ANSWER KEY · DO NOT SHOW UNTIL TASKS A–B ARE DONE

Tasks A and B

Task A · trial-like, DYSP off

- Peak TDD = 160 mg
- Nearer regimen = 150 mg QD
- CUI at 150 mg = 0.688
- CUI at 400 mg = 0.617
- 200 mg BID should not win: DYSP is worth nothing and BMD is expensive.

Task B · dyspareunia priority

- Peak TDD = 400 mg
- Nearer regimen = 200 mg BID
- Argmax moved from Task A? Yes
- Same dummy E–R. One weight changed.
- This is Winzenborg Figure 2b.

ANSWER KEY

Tasks C and D

Task C · DCE average

- Peak TDD = 240 mg (paper ~230 mg)
- Approved dose? No
- Closer marketed-like dose: 150 mg QD
- Honest answer: hypothetical in-between dose
- Hot-flush weight keeps the peak off 400 mg.

Task D · switch the form

- DCE additive = 240 mg
- DCE multiplicative = 310 mg (moves right)
- Bone-sparing additive = 140 mg
- Bone-sparing multiplicative = 170 mg
- Product is not 'always a lower dose'.

ANSWER KEY

Optional virtual patients and the take-home line

Pain-dominant peak = 400 mg (200 mg BID). Bone-sparing peak = 140 mg (150 mg QD).

Under trial-like weights the dummy CUI peaks at 150 mg QD; if dyspareunia is twice as important the peak moves to 200 mg BID; DCE-average preferences peak at a non-approved in-between dose. Weights are the model.